

```

    }
    if (j == n) ans=bip;
  }
  ans *= i0(x)/bi;           Normalize with bessj0.
  return x < 0.0 && (n & 1) ? -ans : ans;
}
}

```

The function `ldexp`, used above, is a standard C and C++ library function for scaling the binary exponent of a number.

CITED REFERENCES AND FURTHER READING:

- Abramowitz, M., and Stegun, I.A. 1964, *Handbook of Mathematical Functions* (Washington: National Bureau of Standards); reprinted 1968 (New York: Dover); online at <http://www.nist.gov/aands>, Chapter 9.
- Carrier, G.F., Krook, M. and Pearson, C.E. 1966, *Functions of a Complex Variable* (New York: McGraw-Hill), pp. 220ff.
- SPECFUN, 2007+, at <http://www.netlib.org/specfun>. [1]
- Hart, J.F., et al. 1968, *Computer Approximations* (New York: Wiley), §6.8, p. 141. [2]
- Numerical Recipes Software 2007, "Coefficients Used in the Bessjy and Bessik Objects," *Numerical Recipes Webnote No. 7*, at <http://www.nr.com/webnotes?7> [3]

6.6 Bessel Functions of Fractional Order, Airy Functions, Spherical Bessel Functions

Many algorithms have been proposed for computing Bessel functions of fractional order numerically. Most of them are, in fact, not very good in practice. The routines given here are rather complicated, but they can be recommended wholeheartedly.

6.6.1 Ordinary Bessel Functions

The basic idea is *Steed's method*, which was originally developed [1] for Coulomb wave functions. The method calculates J_ν , J'_ν , Y_ν , and Y'_ν simultaneously, and so involves four relations among these functions. Three of the relations come from two continued fractions, one of which is complex. The fourth is provided by the Wronskian relation

$$W \equiv J_\nu Y'_\nu - Y_\nu J'_\nu = \frac{2}{\pi x} \quad (6.6.1)$$

The first continued fraction, CF1, is defined by

$$\begin{aligned} f_\nu &\equiv \frac{J'_\nu}{J_\nu} = \frac{\nu}{x} - \frac{J_{\nu+1}}{J_\nu} \\ &= \frac{\nu}{x} - \frac{1}{2(\nu+1)/x - \frac{1}{2(\nu+2)/x - \cdots}} \end{aligned} \quad (6.6.2)$$

You can easily derive it from the three-term recurrence relation for Bessel functions: Start with equation (6.5.6) and use equation (5.4.18). Forward evaluation of the continued fraction by one of the methods of §5.2 is essentially equivalent to backward recurrence of the recurrence relation. The rate of convergence of CF1 is determined by the position of the *turning point* $x_{tp} = \sqrt{\nu(\nu+1)} \approx \nu$, beyond which the Bessel functions become oscillatory. If $x \lesssim x_{tp}$,

convergence is very rapid. If $x \gtrsim x_{\text{tp}}$, then each iteration of the continued fraction effectively increases ν by one until $x \lesssim x_{\text{tp}}$; thereafter rapid convergence sets in. Thus the number of iterations of CF1 is of order x for large x . In the routine `bessel.jy` we set the maximum allowed number of iterations to 10,000. For larger x , you can use the usual asymptotic expressions for Bessel functions.

One can show that the sign of J_ν is the same as the sign of the denominator of CF1 once it has converged.

The complex continued fraction CF2 is defined by

$$p + iq \equiv \frac{J'_\nu + iY'_\nu}{J_\nu + iY_\nu} = -\frac{1}{2x} + i + \frac{i}{x} \frac{(1/2)^2 - \nu^2}{2(x+i) +} \frac{(3/2)^2 - \nu^2}{2(x+2i) +} \dots \quad (6.6.3)$$

(We sketch the derivation of CF2 in the analogous case of modified Bessel functions in the next subsection.) This continued fraction converges rapidly for $x \gtrsim x_{\text{tp}}$, while convergence fails as $x \rightarrow 0$. We have to adopt a special method for small x , which we describe below. For x not too small, we can ensure that $x \gtrsim x_{\text{tp}}$ by a stable recurrence of J_ν and J'_ν downward to a value $\nu = \mu \lesssim x$, thus yielding the ratio f_μ at this lower value of ν . This is the stable direction for the recurrence relation. The initial values for the recurrence are

$$J_\nu = \text{arbitrary}, \quad J'_\nu = f_\nu J_\nu, \quad (6.6.4)$$

with the sign of the arbitrary initial value of J_ν chosen to be the sign of the denominator of CF1. Choosing the initial value of J_ν very small minimizes the possibility of overflow during the recurrence. The recurrence relations are

$$\begin{aligned} J_{\nu-1} &= \frac{\nu}{x} J_\nu + J'_\nu \\ J'_{\nu-1} &= \frac{\nu-1}{x} J_{\nu-1} - J_\nu \end{aligned} \quad (6.6.5)$$

Once CF2 has been evaluated at $\nu = \mu$, then with the Wronskian (6.6.1) we have enough relations to solve for all four quantities. The formulas are simplified by introducing the quantity

$$\gamma \equiv \frac{p - f_\mu}{q} \quad (6.6.6)$$

Then

$$J_\mu = \pm \left(\frac{W}{q + \gamma(p - f_\mu)} \right)^{1/2} \quad (6.6.7)$$

$$J'_\mu = f_\mu J_\mu \quad (6.6.8)$$

$$Y_\mu = \gamma J_\mu \quad (6.6.9)$$

$$Y'_\mu = Y_\mu \left(p + \frac{q}{\gamma} \right) \quad (6.6.10)$$

The sign of J_μ in (6.6.7) is chosen to be the same as the sign of the initial J_ν in (6.6.4).

Once all four functions have been determined at the value $\nu = \mu$, we can find them at the original value of ν . For J_ν and J'_ν , simply scale the values in (6.6.4) by the ratio of (6.6.7) to the value found after applying the recurrence (6.6.5). The quantities Y_ν and Y'_ν can be found by starting with the values in (6.6.9) and (6.6.10) and using the stable upward recurrence

$$Y_{\nu+1} = \frac{2\nu}{x} Y_\nu - Y_{\nu-1} \quad (6.6.11)$$

together with the relation

$$Y'_\nu = \frac{\nu}{x} Y_\nu - Y_{\nu+1} \quad (6.6.12)$$

Now turn to the case of small x , when CF2 is not suitable. Temme [2] has given a good method of evaluating Y_ν and $Y_{\nu+1}$, and hence Y'_ν from (6.6.12), by series expansions that